# Free multi-domain diffusion models and the thermal history of Win-06-01b goethite from the water-saturated step heat HT-Win-06-01b

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**Abstract**

We fit the $^3$He release of the water-saturated step-heating experiment on proton-irradiated Win-06-01b goethite with multi-domain diffusion (MDD) models of $N = 1$–$10$ spherical domains whose $^3$He fractions and sizes are all free, following Farley and Flowers (2012). Each domain structure is then used to fit a burial-and-exhumation thermal history to the bulk-normalized $^4$He/$^3$He spectrum and to predict the (U–Th)/He age. The domain structure stabilizes at $N = 4$. About 97% of the $^3$He resides in one domain slightly larger than the Stage 1 reference domain ($r/r_0 = 1.12$). Three small domains holding 2.8% in total account for the high early diffusivities and the gradual approach to the Stage 1 line. Larger $N$ only duplicate the dominant domain. With this structure, burial paths peak at 58–64 °C between ∼60 and 100 Ma, consistent with regional apatite fission-track constraints. The He age requires crystallization at or after ∼195 Ma.

## 1 Motivation and data

Win-06-01b is dense colloform goethite collected as float in the Flinders Ranges, South Australia, with a (U–Th)/He age of 130–180 Ma (Waltenberg, 2012). Apatite fission-track studies in the region record cooling from roughly 50–70 °C after the mid-Cretaceous (Gibson and Stüwe, 2000; Mitchell et al., 1998), and Triassic intramontane basins less than 1.5 km thick indicate that parts of the ranges were near the surface in the Early Mesozoic (Mitchell et al., 1998). A proton-irradiated aliquot was step-heated in a sealed silica ampoule under water-saturated conditions (HT-Win-06-01b; 34 steps, 170–250 °C), releasing 14.8% of its $^3$He before final fusion. The Stage 1 regression of that experiment defines the reference Arrhenius line used here ($E_a = 157.1$ kJ mol$^{-1}$, $\ln(D_0/a_0^2) = 16.16$ with $D_0/a_0^2$ in s$^{-1}$).

Earlier versions of this analysis split the sample into a high-retentivity domain pinned to the Stage 1 line and a low-retentivity assemblage holding at most 7.5% of the $^3$He. Here that distinction is dropped: all domains are equivalent, and each has a free $^3$He fraction and size.

## 2 Methods

### 2.1 Domain model

The sample is represented by $N$ non-interacting spherical domains that share the Stage 1 activation energy and frequency factor. Domain $j$ holds a fraction $f_j$ of the uniformly distributed, proton-induced $^3$He ($\sum_j f_j = 1$) and has radius $r_j$ relative to the reference radius $r_0$ implied by the Stage 1

regression, so that

$$\left(\frac{D}{a^2}\right)_j = \left(\frac{r_j}{r_0}\right)^{-2} \frac{D_0}{a_0^2} \exp\!\left(-\frac{E_a}{RT}\right). \tag{1}$$

A model with $N$ domains has $k = 2N-1$ free parameters. Sizes are bounded by $10^{-6} \le r_j/r_0 \le 10^3$.

### 2.2 Laboratory release of $^3$He

For a heating schedule of steps $i$ with temperatures $T_i$ and durations $\Delta t_i$, the reference domain accumulates the non-dimensional time

$$\tau_{0,i} = \sum_{m \le i} \frac{D_0}{a_0^2} \exp\!\left(-\frac{E_a}{RT_m}\right) \Delta t_m, \tag{2}$$

and domain $j$ reaches $\tau_{0,i}(r_j/r_0)^{-2}$. An initially uniform sphere releases the cumulative fraction (Crank, 1979)

$$F(\tau) = 1 - \frac{6}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{k^2} \exp\!\left(-k^2\pi^2\tau\right), \tag{3}$$

so the model cumulative and fractional $^3$He yields of the sample are

$$\Sigma F_i = \sum_{j=1}^{N} f_j \, F\!\left(\tau_{0,i}\,(r_j/r_0)^{-2}\right), \qquad \Delta F_i = \Sigma F_i - \Sigma F_{i-1}. \tag{4}$$

### 2.3 Fitting the domain structure to $^3$He

Following Farley and Flowers (2012, supplement A2), the fractional and cumulative yields are weighted equally, and $\{f_j, r_j\}$ minimize

$$S = \sum_{i=1}^{34} \left(\Delta F_i^{\mathrm{mod}} - \Delta F_i^{\mathrm{obs}}\right)^2 + \sum_{i=1}^{34} \left(\Sigma F_i^{\mathrm{mod}} - \Sigma F_i^{\mathrm{obs}}\right)^2, \tag{5}$$

with yields expressed as fractions of the total $^3$He, including the fusion step. Models are built up one domain at a time: the $N$-domain search starts from the $(N-1)$-domain solution plus one new domain trialled at 15 sizes and two fractions, together with ten random starts. The fit therefore cannot get worse as $N$ increases. To compare models of different complexity, we use the Bayesian information criterion for Gaussian residuals of unknown variance,

$$\mathrm{BIC} = n \ln(S/n) + k \ln n, \qquad n = 68. \tag{6}$$

The Arrhenius and delta plots are not fitted. They follow from applying the single-sphere inversion of Fechtig and Kalbitzer (1966) to the model cumulative release. Their misfit is reported as a reduced $\chi^2$ against the bounding-method uncertainties in $\ln(D/a^2)$.

### 2.4 Radiogenic $^4$He in nature

Each domain produces $^4$He uniformly at rate $P$ and loses it by diffusion with a zero-concentration boundary,

$$\frac{\partial C}{\partial t} = \frac{D}{a^2} \frac{1}{x^2} \frac{\partial}{\partial x}\left(x^2 \frac{\partial C}{\partial x}\right) + P, \qquad C(1,t) = 0, \quad C(x, t_{\mathrm{form}}) = 0, \tag{7}$$

where $x$ is radial position normalized to the domain radius and $t_{\rm form}$ is the crystallization age. With $C = \sum_k A_k(t)\sin(k\pi x)/x$ (Carslaw and Jaeger, 1959), each mode evolves as

$$\frac{\mathrm{d}A_k}{\mathrm{d}t} = -k^2\pi^2\frac{D}{a^2}A_k + c_k P, \qquad c_k = \frac{2(-1)^{k+1}}{k\pi}, \tag{8}$$

which over an interval $\Delta t$ (2 Myr) at constant temperature has the exact solution

$$A_k \leftarrow A_k e^{-\lambda_k} + c_k P\,\Delta t\,\frac{1-e^{-\lambda_k}}{\lambda_k}, \qquad \lambda_k = k^2\pi^2\frac{D}{a^2}\Delta t. \tag{9}$$

We use 8000 modes. The $^4$He content per unit volume of domain $j$ is $N_j = 3\sum_k A_{k,j}(-1)^{k+1}/(k\pi)$. Taking U and Th to be distributed like the $^3$He and neglecting parent decay ($\sim$3% over 200 Myr), the bulk He age is

$$t_{\rm He} = \frac{1}{P}\sum_j f_j\,N_j. \tag{10}$$

### 2.5 Laboratory degassing of the natural profiles

In the laboratory each mode decays as $\exp(-k^2\pi^2\tau)$. The reference times $\tau_{0,i}$ used for the $^4$He prediction are solved from the *measured* cumulative $^3$He by inverting Eq. 4. The predicted spectrum is therefore independent of the heating schedule and of step-to-step kinetic scatter:

$$\frac{R_{{\rm step},i}}{R_{\rm bulk}} = \frac{\Delta N_{4,i}/N_{4,\rm tot}}{\Delta F_i^{\rm obs}}, \qquad \Delta N_{4,i} = \sum_j f_j\left[N_j(\tau_{0,i-1}) - N_j(\tau_{0,i})\right], \tag{11}$$

where $N_j(\tau)$ is the $^4$He remaining in domain $j$ after laboratory time $\tau(r_j/r_0)^{-2}$ and $N_{4,\rm tot} = \sum_j f_j N_j(0)$ includes the gas released at fusion, consistent with the measured bulk ratio.

### 2.6 Thermal histories

The thermal history is piecewise linear through $(t_{\rm form}, T_s)$, $(t_{\rm peak}, T_{\rm max})$, (50 Ma, $T_s$) and $(0, T_s)$ with $T_s = 20\,^\circ$C. The goethite forms at the surface, is buried and heated to a peak temperature, and is exhumed by the Paleogene, when Paleogene sediments in Flinders Ranges valleys show that basement was exposed (Mitchell et al., 1998). For each domain structure and each $t_{\rm form} \in \{180, 190, 200, 210\}$ Ma, $(t_{\rm peak}, T_{\rm max})$ are optimized by Nelder–Mead against

$$\chi^2 = \sum_i \left(\frac{R_i^{\rm mod} - R_i^{\rm M}}{\sigma_i}\right)^2, \qquad \sigma_i = \max\left[\tfrac{1}{2}\left(R_i^{\rm UB} - R_i^{\rm LB}\right), 0.02\right], \tag{12}$$

over the 24 steps that have lag-correction bounds and more than 0.1 pccSTP of $^3$He. M, LB and UB are the central, lower-bound and upper-bound bulk-normalized ratios of the bounding method. We also map $\chi^2$ on a grid of $t_{\rm peak}$ (2.5 Myr steps) and $T_{\rm max}$ (1 °C steps) for $t_{\rm form}$ = 180, 195 and 210 Ma. The joint $1\sigma$ region is $\chi^2 \le \chi^2_{\rm min} + 3.53\max(1, \chi^2_\nu)$, and the predicted He age is contoured at the limits of the measured range. For each crystallization age, we draw 300 burial paths from the gridded posterior $p \propto \exp[-(\chi^2 - \chi^2_{\rm min})/(2s)]$, with $s = \max(1, \chi^2_\nu)$, jittering each draw within its grid cell. With two free parameters on a dense grid, this is equivalent to Markov-chain Monte Carlo sampling of the posterior.

# 3 Results

## 3.1 Domain structure from $^3$He

Table 1 lists the fits for all $N$, and Fig. 1 shows the fitted domain spectra.

- $N = 1$**.** A single domain cannot fit the release (Arrhenius $\chi^2_\nu = 23$).
- $N = 2$**.** The second domain captures the high early diffusivities but leaves a poor Arrhenius fit ($\chi^2_\nu = 3.2$).
- $N = 3$**.** The yield misfit falls by more than half and the Arrhenius $\chi^2_\nu$ falls to 1.6.
- $N = 4$**.** The smallest domain splits in two, and the fit improves slightly further (Arrhenius $\chi^2_\nu = 1.58$; BIC indistinguishable from $N = 3$).
- $N \geq 5$**.** Every added domain coincides in size with the dominant domain. The fit is unchanged and BIC rises.

We therefore adopt $N = 4$ as the complete description of the $^3$He data. Its domains are:

- 0.54% of the $^3$He at $r/r_0 = 6.6 \times 10^{-4}$;
- 0.66% at $2.6 \times 10^{-3}$;
- 1.6% at 0.033;
- 97.2% at 1.12.

The three small domains are fully degassed during the experiment. The dominant domain releases 12% of its $^3$He, at a diffusivity 0.80 times the Stage 1 line. The Stage 1 line is therefore a slightly fast description of the dominant domain: it was regressed on bulk-inverted diffusivities that still carry a small contribution from the fast domains. With the dominant domain free to differ from the Stage 1 reference, the model follows the delta plot closely, including the gradual decline between $\Sigma F_{^3\mathrm{He}} = 0.04$ and 0.07 (Fig. 2). Only the prograde 240 and 250 °C steps in that interval lie above the model, by ∼1–1.7$\sigma$. Figure 6 shows the fitted yields and the resulting Arrhenius and delta plots for $N = 1$–5.

**Table 1:** Summary of the free multi-domain models. RMS is the yield misfit ($\times 10^{-4}$, fraction of total $^3$He); "fast" is the $^3$He fraction in domains with $r/r_0 < 0.1$; $r_{\mathrm{main}}$ is the size of the dominant domain. Thermal columns give the best burial path with a predicted He age of 130–180 Ma; the last two columns give the ranges of $T_{\max}$ (°C) and $t_{\mathrm{peak}}$ (Ma) within the joint 1$\sigma$ region of the $\chi^2$ maps that also satisfy the He age (maps computed for $N \leq 4$).

| $N$ | $k$ | RMS | BIC | Arrh. $\chi^2_\nu$ | fast (%) | $r_{\mathrm{main}}/r_0$ | $^4$He/$^3$He $\chi^2_\nu$ | $t_{\mathrm{form}}$ | $t_{\mathrm{peak}}$ | $T_{\max}$ | $t_{\mathrm{He}}$ | $T_{\max}$ (1$\sigma$) | $t_{\mathrm{peak}}$ (1$\sigma$) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 84.06 | -645.7 | 23.33 | 0.0 | 0.79 | 6.61 | 200 | 55.0 | 59.5 | 179 | 60–60 | 55.0–55.0 |
| 2 | 3 | 19.76 | -834.2 | 3.24 | 2.1 | 1.03 | 1.33 | 200 | 97.8 | 65.0 | 178 | 63–67 | 87.5–100.0 |
| 3 | 5 | 8.50 | -940.5 | 1.60 | 2.8 | 1.12 | 5.87 | 190 | 74.0 | 60.4 | 173 | 57–64 | 57.5–92.5 |
| 4 | 7 | 8.05 | -939.4 | 1.58 | 2.8 | 1.12 | 5.59 | 190 | 74.4 | 60.4 | 173 | 57–64 | 60.0–95.0 |
| 5 | 9 | 8.05 | -930.9 | 1.71 | 2.8 | 1.12 | 5.59 | 190 | 74.4 | 60.4 | 173 | – | – |
| 6 | 11 | 8.05 | -922.5 | 1.85 | 2.8 | 1.12 | 5.59 | 190 | 74.4 | 60.4 | 173 | – | – |
| 7 | 13 | 8.05 | -914.1 | 2.03 | 2.8 | 1.12 | 5.59 | 190 | 74.4 | 60.4 | 173 | – | – |
| 8 | 15 | 8.05 | -905.6 | 2.24 | 2.8 | 1.12 | 5.59 | 190 | 74.4 | 60.4 | 173 | – | – |
| 9 | 17 | 8.05 | -897.2 | 2.51 | 2.8 | 1.12 | 5.59 | 190 | 74.4 | 60.4 | 173 | – | – |
| 10 | 19 | 8.05 | -888.7 | 2.84 | 2.8 | 1.12 | 5.59 | 190 | 74.4 | 60.4 | 173 | – | – |

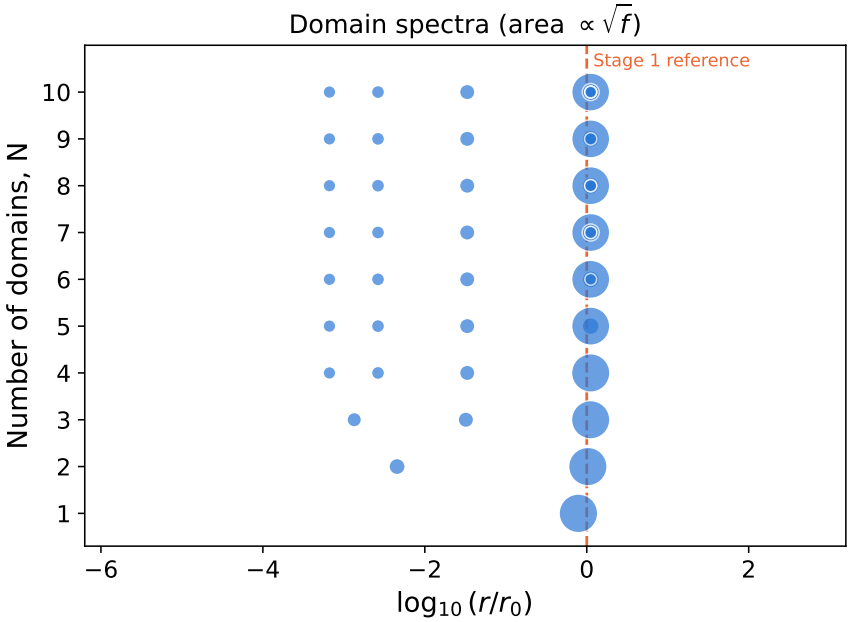



**Figure 1:** Domain spectra of the best-fitting models. Symbol area scales with the square root of the $^3$He fraction; the dashed line marks the Stage 1 reference domain. The three small domains are unchanged from $N = 4$ onward; domains added beyond $N = 4$ coincide in size with the dominant domain.

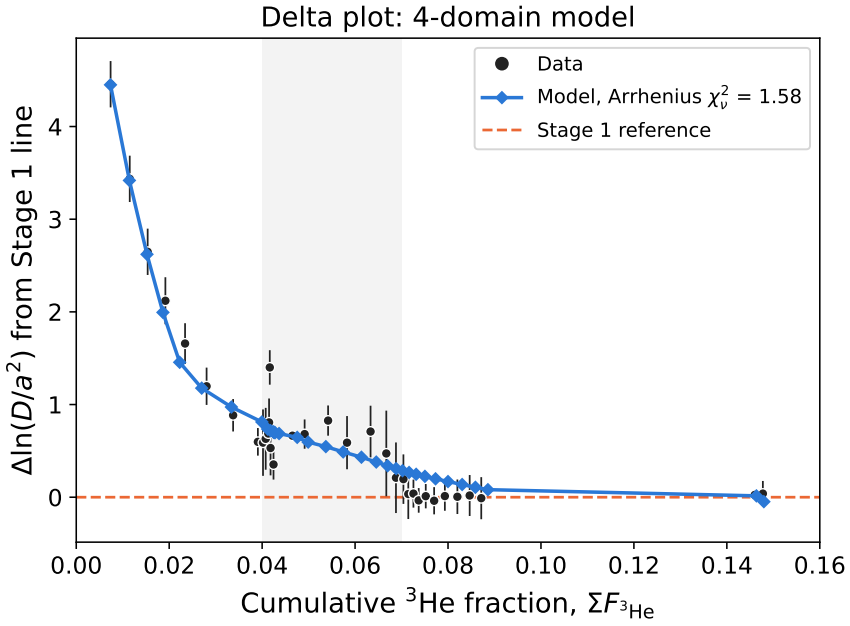



**Figure 2:** Delta plot of the measured $^3$He diffusivities (departure from the Stage 1 line) and the curve implied by the four-domain fit to the $^3$He yields (the delta plot itself is not fitted). The shaded band marks $\Sigma F_{^3\mathrm{He}} = 0.04$–$0.07$, where the prograde 220–250 °C steps show elevated diffusivities.

### 3.2 Thermal history from $^4$He/$^3$He

With the four-domain structure fixed, burial histories peaking at ≈60 °C fit the $^4$He/$^3$He spectrum best (Figs. 3 and 4). The spectrum constrains the peak temperature tightly, while peak time trades off against crystallization age (Table 2):

- **180 Ma.** Posterior paths peak at 59–63 °C between 77.5 and 60 Ma.
- **195 Ma.** Paths peak at 59–64 °C between 92.5 and 72.5 Ma.
- **210 Ma.** Paths peak at 58–64 °C between 102.5 and 80 Ma.

The He-age isochrons then select among these. All posterior paths for crystallization at 180 Ma and 97% of those at 195 Ma predict He ages within the measured 130–180 Ma range. None do for crystallization at 210 Ma, where every path predicts an age older than 180 Ma. The surviving paths peak in the Late Cretaceous, within or just after the apatite fission-track window. Their peak temperatures fall within the 50–70 °C range of Gibson and Stüwe (2000) and at the <60 °C limit reported near Copley (Mitchell et al., 1998).

The absolute quality of the spectrum fit is modest ($\chi^2_\nu \approx 5.2$–$5.8$). The model underpredicts the ratios between $\Sigma F_{^3\mathrm{He}} = 0.04$ and 0.07 and overpredicts the long 250 °C step. A two-domain structure fits the spectrum considerably better ($\chi^2_\nu \approx 1.3$; peak ≈65 °C at 80–100 Ma). However, it reproduces the $^3$He Arrhenius array poorly (Fig. 6), so we do not adopt it.

**Table 2:** Posterior burial paths for the four-domain structure (300 samples per crystallization age; 16th–84th percentiles).

| $t_{\mathrm{form}}$ (Ma) | best $\chi^2_\nu$ | $T_{\mathrm{max}}$ (°C) | $t_{\mathrm{peak}}$ (Ma) | He age 130–180 Ma |
|---|---|---|---|---|
| 180 | 5.81 | 59–63 | 60.0–77.5 | 100% |
| 195 | 5.50 | 59–64 | 72.5–92.5 | 97% |
| 210 | 5.20 | 58–64 | 80.0–102.5 | 0% |

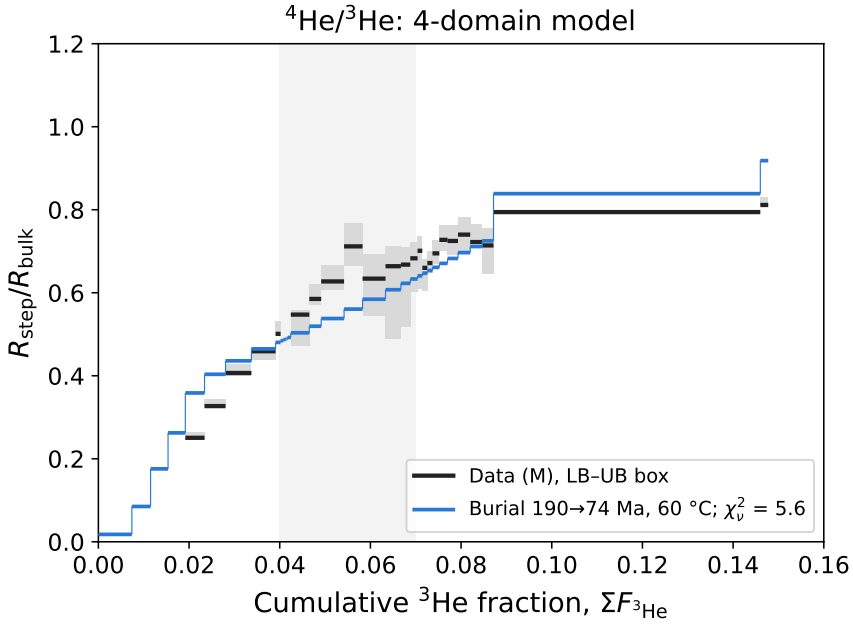



**Figure 3:** Bulk-normalized $^4$He/$^3$He spectrum (black, bounding-method LB–UB boxes) and the prediction of the best age-consistent burial path for the four-domain structure.

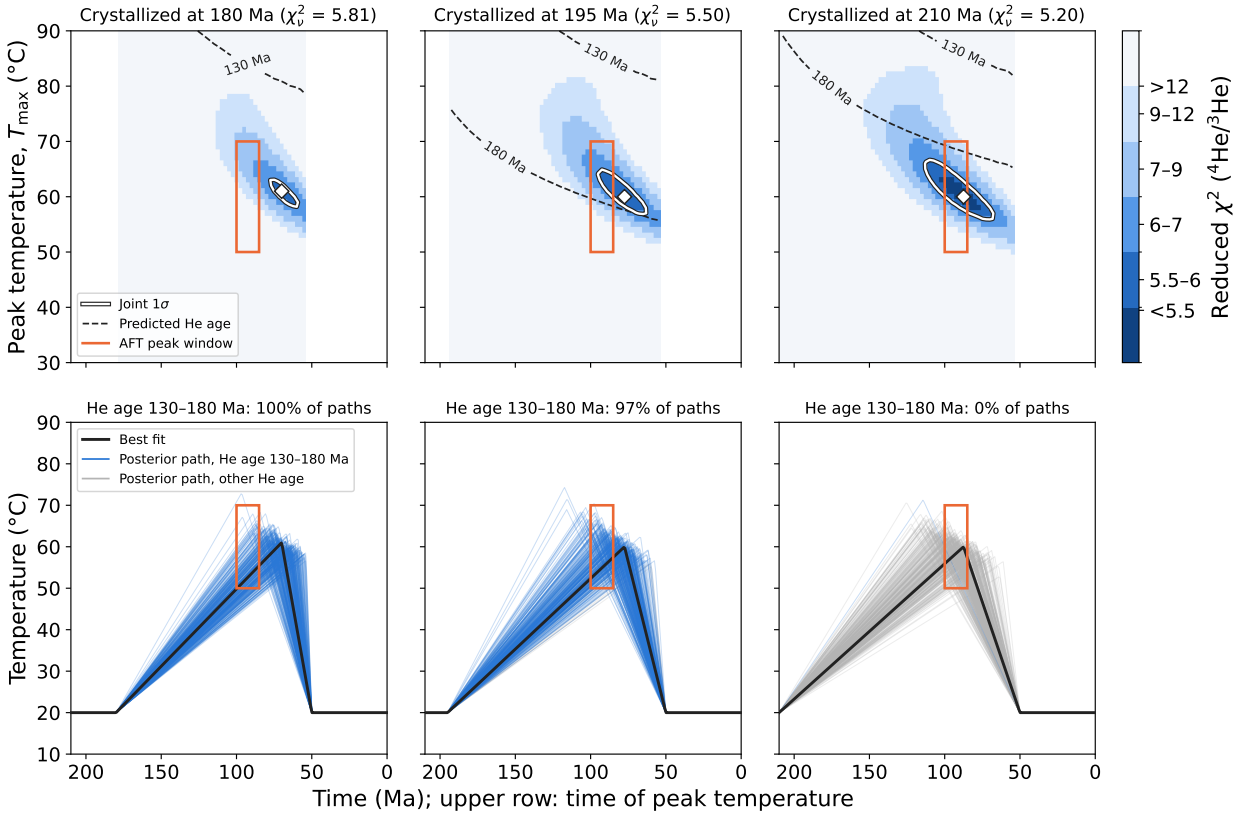



**Figure 4:** Thermal-history constraints for the four-domain structure, with one column per assumed crystallization age (180, 195 and 210 Ma). Top row: reduced $\chi^2$ of the $^4$He/$^3$He spectrum as a function of the time and magnitude of peak temperature. The colour scale is anchored to this model's minimum. White outlines show the joint $1\sigma$ region, dashed lines the predicted bulk He age at 130 and 180 Ma, diamonds the best fit, and orange boxes the apatite fission-track peak window (Gibson and Stüwe, 2000; Mitchell et al., 1998). Bottom row: 300 burial paths drawn from the gridded posterior for each crystallization age. Blue paths predict He ages within 130–180 Ma; grey paths do not. The best fit is shown in black.

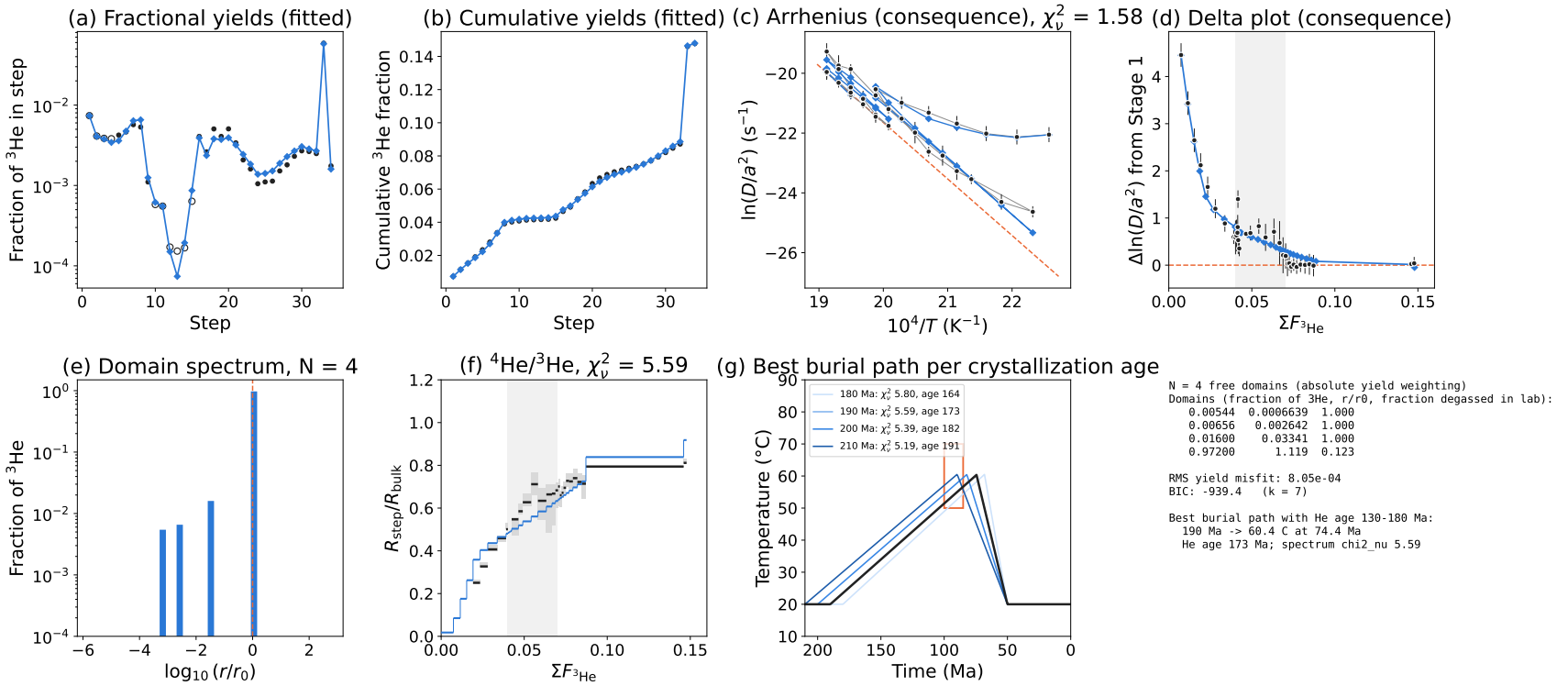



**Figure 5:** Diagnostic summary of the four-domain model: (a) fitted fractional and (b) cumulative $^3$He yields; (c) Arrhenius and (d) delta plots implied by the fit; (e) domain spectrum; (f) $^4$He/$^3$He spectrum with the best age-consistent burial path; (g) best path for each crystallization age; (h) parameter summary, including the fraction of each domain degassed during the experiment.

## 4 Discussion

**Robust conclusions.** Across all domain structures, the dominant domain holds ≈97% of the $^3$He and degasses slightly more slowly than the Stage 1 line. Fast domains hold only 2–3%. With the four-domain structure, burial paths peak at 58–64 °C in the Late Cretaceous, followed by exhumation, and the He age requires crystallization at or after ∼195 Ma. This thermal history is consistent with the apatite fission-track record of 50–70 °C before cooling after ∼85 Ma (Gibson and Stüwe, 2000) and with the shallow Triassic basins and low-rank coals of the Flinders Ranges (Mitchell et al., 1998).

**Remaining misfit.** The four-domain structure leaves systematic misfit in the $^4$He/$^3$He spectrum. Its intermediate domain ($r/r_0 \approx 0.03$, 1.6% of the $^3$He) degasses at $\Sigma F_{^3\mathrm{He}} \approx 0.02$–$0.06$. With the common activation energy, it would have lost most of its radiogenic $^4$He in nature. That depresses the predicted ratios where the measured ratios are highest. Diffusion-only remedies that preserve the $^3$He fit are:

- a higher activation energy for the intermediate domain, which the retrograde segments of the experiment could test;
- domain-dependent $^4$He production, for example from U–Th zoning across the colloform bands.

Each adds one parameter. A finite non-diffusive release during the first heating to 220–250 °C remains possible but is no longer needed to explain the $^3$He data.

**Caveats.**

- The temperature scale rests on extrapolating the Stage 1 activation energy from 200–250 °C to near-surface temperatures (± ∼4 °C near 60 °C, not propagated here).
- The exhumation time (50 Ma) and surface temperature (20 °C) are fixed.
- Domains that barely degas in the experiment are constrained only through their summed fraction.

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# A Supplementary figure

Diagnostic figures and $\chi^2$ maps for the other domain structures are written to `Figures/generated/mdd_free/abso`

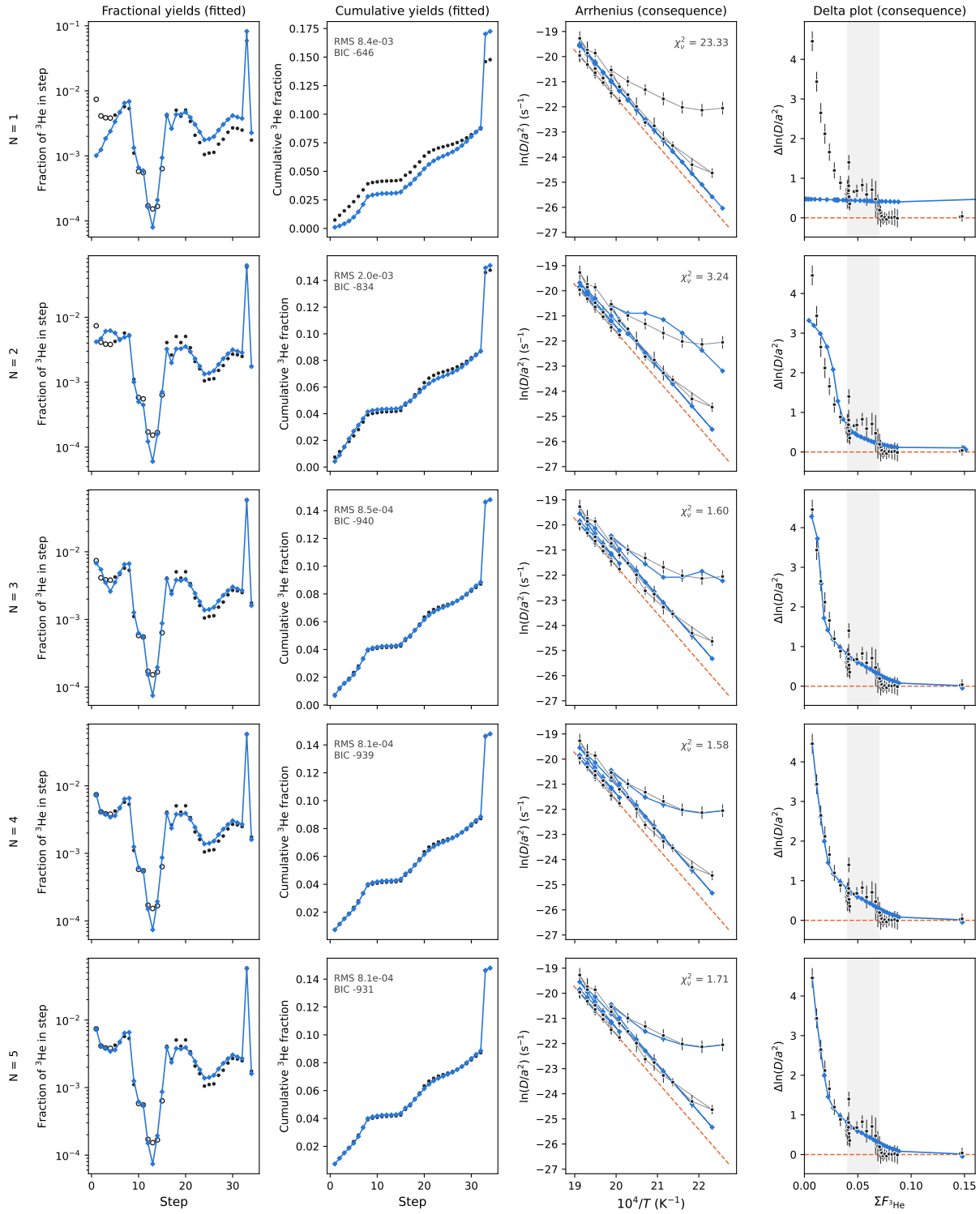



**Figure 6:** Fits to the $^3$He data for $N = 1$–5 domains (rows). Columns: fitted fractional and cumulative yields, and the Arrhenius and delta plots implied by each fit. Hollow symbols are low-signal steps or steps without lag bounds.